

Wednesday, June 17, 2026

Since you're building the repository as documentation first, I would populate the remaining files with foundational content before writing code.

`governance/principles.md`

EchoAuth Governance Principles

Principle 1: Authority Before Execution

No action may be executed without authorization from the designated authority layer.

Principle 2: Human Authority Preservation

Human authority remains the final source of approval, denial, escalation, and accountability.

Principle 3: Refusal Is Compliant

A refusal to execute is a valid and compliant system state when authorization requirements are not satisfied.

Principle 4: Separation of Powers

Meaning, judgment, authority, and execution shall remain independent functions.

NI-AI interprets.

Governance evaluates.

EchoAuth authorizes.

Execution performs.

Principle 5: Auditability

Every authorization decision must be traceable through logs and decision records.

Principle 6: Escalation Integrity

When confidence, authority, or context is insufficient, the system escalates rather than assumes.

Principle 7: Deterministic Authorization

The same inputs, authorities, and conditions must produce the same authorization outcome.

Principle 8: Accountability Preservation

Multi-party coordination shall never remove responsibility from authorized actors.

Core Statement

Coordination proposes.

Governance permits.

Execution proceeds after authorization.

architecture/system-overview.md

EchoAuth System Overview

Purpose

EchoAuth is a deterministic authorization framework designed to ensure governance always precedes execution.

Architectural Layers

Layer 1: Meaning

Component: NI-AI

Responsibilities:

Interpret information

Analyze context

Generate proposals

Produce recommendations

NI-AI never executes actions.

Layer 2: Judgment

Component: MCG (Multi-Coherence Governance)

Responsibilities:

Evaluate proposals

Assess coherence

Validate safety constraints

Determine governance outcomes

MCG never executes actions.

Layer 3: Authority

Component: EchoAuth

Responsibilities:

Verify authority ownership

Validate permissions

Apply authorization rules

Produce authorization decisions

EchoAuth controls execution access.

Layer 4: Execution

Component: CEG (Controlled Execution Gateway)

Responsibilities:

Execute approved actions

Enforce governance decisions

Record execution outcomes

Execution cannot bypass authorization.

Operational Flow

Information enters NI-AI.

NI-AI generates interpretation.

MCG evaluates interpretation.

EchoAuth validates authority.

Authorization decision is produced.

CEG executes approved actions.

Audit records are generated.

Design Objective

Maintain clear separation between interpretation, governance, authorization, and execution.

patents/patent-summary.md

EchoAuth Patent Summary

Inventor

Nicholas B. Carty

Assignee

HeliosFi LLC

Patent Family

EchoAuth Governance and Authorization Framework

Core Innovation

EchoAuth introduces a deterministic authorization layer that separates:

Meaning

Judgment

Authority

Execution

into independent operational domains.

Key Concepts

Deterministic Authorization

Authorization outcomes are generated from explicit authority ownership and governance rules.

Parent-Anchored Governance

Authority originates from designated human stakeholders and cannot be overridden by autonomous coordination.

Refusal-First Architecture

The system defaults to refusal when authorization conditions are not satisfied.

Governance Before Execution

No action may be executed until governance and authorization requirements are fulfilled.

Multi-Party Accountability

Coordination across organizations and stakeholders preserves responsibility and traceability.

Application Domains

Healthcare

Education

Public Institutions

Caregiver Systems

AI Governance

Identity Management

Multi-Organization Coordination

Long-Term Goal

Establish a universal governance layer for human-supervised intelligent systems.

specs/echoauth-spec.md

EchoAuth Technical Specification

Version

0.1

System Type

Deterministic Authorization Framework

Core Components

NI-AI

Role:

Interpretation Layer

Functions:

Context interpretation

Recommendation generation

Information synthesis

MCG

Role:

Governance Layer

Functions:

Coherence evaluation

Governance assessment

Decision support

EchoAuth

Role:

Authorization Layer

Functions:

Authority verification

Permission validation

Authorization issuance

CEG

Role:

Execution Layer

Functions:

Action execution

Enforcement

Audit logging

Authorization States

APPROVED

Authorization requirements satisfied.

DENIED

Authorization requirements not satisfied.

HOLD

Additional review required.

ESCALATED

Transferred to higher authority.

Security Objectives

Authority integrity

Auditability

Deterministic outcomes

Accountability preservation

Human oversight

System Invariant

No execution without authorization.

These four files will give EchoAuth a professional baseline structure. After that, the next major documents should be:

governance/authority-model.md

architecture/system-diagram.md

patents/claim-index.md

specs/security-model.md

Those four documents are where your MCG, CEG, S23 invariants, PNRE, TLE, and the rest of the deeper EchoAuth architecture should be captured.